C-M-50a
March 9, 1966

SUPERSEDING
Int. Fed. Spec. C-M-0050 (AGR-AMS)
June 22, 1964 and
Fed. Spec. C-M-341a
February 7, 1955

FEDERAL SPECIFICATION

MALTED MILK

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for the preparation and packaging of dry malted milk for use by agencies of the Federal Government.

1.2 Classification. Dry malted milk covered by this specification shall be of the following types and classes, as specified (see 6.3):

1.2.1 *Type.*

I—Natural flavored malted milk.
II—Chocolate flavored malted milk.

1.2.2 *Class.*

1—Instantized.
2—Conventional.
3—Tablet.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

Federal Specifications:

TT-C-495—Coatings, Exterior, for Tinned Food Cans.
PPP-B-636—Box, Fiberboard.
PPP-C-29—Canned Subsistence Items, Packaging and Packing of.
PPP-C-96—Cans, Metal, 28 Gage and Lighter.

Federal Standards:

Fed. Std. No. 101—Preservation, Packaging, and Packing Materials; Test Procedures.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-L-1497—Labeling of Metal Cans for Subsistence Items.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.
MIL-STD-671—Sanitary Standards for Milk Evaporating and Dry Plants.

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Governmental:

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

Working Reference of Livestock Regulatory Establishments, Stations and Officials.

(Copies of the above publications may be obtained upon application, accompanied by money order or cash to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402. Prices may be obtained from the Superintendent of Documents.)

U. S. Standards for Condition of Food Containers.

Minimum Specifications for Approved Plants Manufacturing, Processing and Packaging Dairy Products.

Dairy Plants Surveyed and Approved for USDA Grading Service.

Methods of Laboratory Analyses for Dry Whole Milk, Nonfat Dry Milk, Dry Buttermilk and Dry Whey.

(Single copies may be obtained without charge obtained without charge from the Dairy Division, Consumer and Marketing Service, United States Department of Agriculture, Washington, D. C., 20250.)

Official Grain Standards of the United States.

Single copies may be obtained without charge from the Grain Division, Consumer and Marketing Service, United States Department of Agriculture, Federal Center Building, Hyattsville, Md., 20781.)

Nongovernmental:

Official Methods of Analysis of the Association of Official Agricultural Chemists.

(Published by the Association of Official Analytical Chemists, P. O. Box 540, Benjamin Franklin Station, Washington, D. C., 20044.)

Standard Methods for the Examination of Dairy Products.

(Copies may be obtained upon application to the American Public Health Association, Inc., 1790 Broadway, New York, N. Y., 10019.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 *Milk.* The raw milk used in the preparation of dry malted milk shall be drawn from cows, in herds located in a modified accredited area or from cows in herds fully accredited as tuberculosis-free and from cows in herds located in a modified certified area or from cows in herds certified brucellosis-free by the U. S. Department of Agriculture or in the process of being accredited or certified.

3.1.1.1 *Bacterial and sediment requirements. (Civil agencies.)* The raw milk supply shall conform with the bacterial and sediment requirements for acceptable milk set forth in the Minimum Specifications for Approved Plants Manufacturing, Processing and Packaging Dairy Products.

3.1.1.1.1 Raw milk procured for military use shall conform with the bacterial and sediment requirement defined in MIL-STD-671.

3.1.2 *Other milk products.* Other milk products commonly used in the manufacture of dry malted milk shall be produced from milk complying with 3.1.1. At the time of use, these products shall contain no objectionable flavors and odors.

3.1.3 *Wheat.* The wheat shall be No. 2 or better of any classes of wheat as defined in the Official Grain Standards of the United States for wheat. The wheat shall be clean, sound, and scoured.

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3.1.4 *Farina or wheat flour.* The farina or wheat flour shall conform to the Definitions and Standards of Identity of the Federal Food, Drug, and Cosmetic Act for Cereal Flours and Related Products except that the ash content requirement shall not apply.

3.1.5 *Barley malt or malted wheat.* The barley malt or malted wheat shall be free from root hairs, extraneous material, and signs of insect and rodent contamination. It shall have a clean characteristic flavor.

3.1.6 *Sodium chloride, sodium bicarbonate, and potassium bicarbonate.* These chemicals shall be of food grade quality.

3.1.7 *Cocoa and chocolate liquor.* Cocoa (medium fat) or chocolate liquor or a combination thereof, shall conform to the Definitions and Standards of the Federal Food, Drug, and Cosmetic Act for Chocolate and Cocoa Products.

3.1.8 *Sugar.* Sugar shall be sucrose of food grade quality. The sugar shall be free from visible extraneous materials.

3.1.9 *Flavoring, (type II, optional ingredient).* Vanilla, vanillin, or artificial vanilla flavoring or a combination thereof, shall be of food grade quality and possess a pleasing and desirable flavor.

3.1.10 *Emulsifiers, (class 1, optional ingredient).* Emulsifiers shall be food grade lecithin or other suitable emulsifiers of food grade quality which have been authorized by regulations, promulgated under the Food Additives Amendment to the Federal Food, Drug, and Cosmetic Act for use in dry whole milk. The amount of emulsifiers used shall not exceed 1-1/2 percent of the dry finished product.

3.1.11 *Water.* The water used in the manufacture of dry malted milk under this specification shall be of a safe sanitary quality and approved by State or Federal authorities. It may be subjected to suitable chemical or heat treatment and shall not impart undesirable odors or flavors to the finished product.

3.2 Preparation.

3.2.1 *Processing.* Barley malt, wheat malt, water, and wheat, farina, or wheat flour, shall be combined to secure full enzymatic action. Milk or milk products (or a combination of both) shall be added to the liquid separated from this mash. In addition, sodium chloride, sodium bicarbonate, potassium bicarbonate (or any combination of these) may be added.

3.2.2 *Milk storage.* Unless processing is started within two hours following receipt of the milk at the processing plant, all incoming milk shall be cooled to 45°F. or lower until processed. In no case shall the milk be held in the plant longer than 48 hours prior to processing. Milk held longer than one hour between stages of processing shall be held at a temperature of above 135°F. or below 45°F., but in no instance shall the milk be held at 135°F. or above for longer than two hours.

3.2.3 *Pasteurization.* The raw milk or raw milk and wort mixture shall be pasteurized prior to condensing so that every particle will be heated to a temperature of not less than 145°F., and held continuously at such temperature for not less than 30 minutes or heated to a temperature of not less than 161°F. and held at such temperature for not less than 15 seconds, or any other temperature and time treatments approved by State or Federal authorities.

3.2.4 *Drying.* Class 1 product may be produced by partially rehydrating and redrying conventional dry malted milk in order to improve its dispersing characteristics.

3.3 Finished product.

3.3.1 *Physical requirements.*

3.3.1.1 *Flavor.* Type I product shall have a pleasing, pronounced natural malt flavor. Type II product shall have a pleasing, pro-

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nounced chocolate and malt flavor. Both types shall be free from definite scorched, or slight rancid, slight stale, slight bitter, slight neutralizer, or other objectionable flavors or odors.

3.3.1.2 *Color.* Type I product shall be natural tan to golden yellow. Type II product shall be light to medium chocolate color.

3.3.1.3 *Appearance and texture.* The finished product shall be in powder (class 1—instantized, or class 2—conventional) or tablet form (class 3). If in powder form, it shall have a uniform color and appearance, free from foreign material and lumps of product that do not fall apart under slight pressure. If in tablet form, it shall be pressed into tablets approximately 5/8 inch in diameter and 1/4 inch in thickness—food grade hydrogenated oil may be added up to 0.5 percent by weight to facilitate compression.

3.3.2 *Analytical requirements.* The finished product shall conform to the following analytical requirements:

	Type I	Type II
Milk fat (not less than)	7.5 percent	2.8 percent
Protein (not less than)	12.5 percent	6.0 percent
Moisture (not more than)	3.5 percent	3.5 percent
Phosphatase activity (not more than)	Both types 4 micrograms of phenol per ml. of reconstituted product	
Bacterial estimate by standard plate count (not more than)	30,000 per g.	30,000 per g.
Coliform count (not more than)	10 per g.	10 per g.
Oxygen in headspace gas (as specified) (not more than)	2.0 percent	2.0 percent

3.3.2.1 *Class 1 product (instantized).* In addition to analytical requirements of 3.3.2 for conventional products, class 1 product shall comply with the following analytical requirements:

3.3.2.1.1 *Dispersibility performance.* Mininum dispersibility performance for type I—40.0 g. and for type II—44.0 g. when tested as specified in 4.3.3.

3.3.2.1.2 *Density.* The density shall be such that one liter of the dry product shall weigh not less than 490 g. when tested as specified in 4.3.4.

3.4 **Age of product.** The finished product shall be shipped within 90 days following the initial drying of the product, unless otherwise specified by the procuring agency (see 6.3).

3.5 **Plant, equipment, workmanship.** The receiving stations, processing plant, premises, equipment, operations, personnel practices, and sanitary practices used in the production and transportation of the milk shall meet the applicable facilities and sanitation requirements in the Minimum Specifications for Approved Plants Manufacturing, Processing and Packaging Dairy Products or when procurements are for military use, MIL-STD-671. The milk shall be prepared and processed with minimum delay between the various steps in production.

3.6 **Federal Food, Drug, and Cosmetic Act.** All deliveries shall conform in every respect to the provision of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** The supplier is responsible for meeting all the provisions of the specification prior to submission of supplies to the Government for acceptance. Unless otherwise specified in the contract or order (see 6.3), the supplier is required to perform the examination and tests prescribed herein using his own or any other inspection facilities and services acceptable to the Government.

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4.1.1 Inspection records of the examination and tests shall be complete and made available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.2 *Sampling.* Samples of finished products covered by this specification and samples of any materials, components or constituents entering products covered by this specification may be taken at any time by the Government inspector to determine compliance with the requirements. Such samples as are required shall be at the expense of the contractor.

4.2 Quality conformance inspections.

4.2.1 *General.* Inspection shall be made by the procuring agency, or a duly authorized representative, at the time and place designated by the procuring agency. It may be made at the site of manufacture, both during and after processing, at any point in transit, and/or after delivery at the point of destination.

4.2.2 *Preacceptance factory sanitation inspection.*

4.2.2.1 *For civil agencies.* Award of contracts shall be limited to plants which have received prior sanitary approval by the procuring agency or a duly authorized representative. Plants listed or eligible for listing in the current bulletin Dairy Plants Surveyed and Approved for USDA Grading Service may be accepted as evidence of compliance.

4.2.2.2 *For military agencies.* Award of contracts shall be limited to establishments which have received prior sanitary approval by the Military Veterinary Services in accordance with MIL-STD-671.

4.2.3 *Factory inspection.* The passing as satisfactory of any detail of processing or materials shall not relieve the contractor of responsibility for faulty workmanship or materials which may be discovered at any time prior to final acceptance.

4.2.4 *Final inspection.* Unless otherwise specified (see 6.3), final inspection shall be made after delivery at the point of destination.

4.2.5 *USDA Certificate of Quality and Inspection.* When specified by the procuring agency (see 6.3), the contractor shall furnish a U. S. Department of Agriculture inspection or grading certificate evidencing quality, quantity and condition, as outlined in procurement documents. (See 6.3 for note on small lot orders.)

4.2.6 *Basis for acceptance by the Government.* The contractor is responsible for offering to the Government for acceptance only those items that conform to all contractural requirements. The Government will conduct quality conformance inspections required to determine acceptability of the components and finished product in accordance with inspection procedures set forth in 4.2.7 through 4.3.4 as follows:

4.2.6.1 Verify contractor's inspection and test records (see 4.2.7, 4.2.8, and 4.3.2.1) as to compliance with the specification.

4.2.6.2 Select samples of the finished product and perform verification inspections as directed by procuring agency. (See 4.2.9.)

4.2.7 *Inspection of components (contractor inspections).*

4.2.7.1 *Milk.* Records of the results of the bacterial and sediment examination of the milk supply shall be available for review by the procuring agency. If upon examination of the records it is determined that the processing plant has processed raw milk not in compliance with the requirements of 3.1.1, the finished product shall be rejected. Failure to maintain records as required shall be basis for rejection of the finished product.

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4.2.7.1.1 Inspection shall be made to assure that milk used in the preparation of this product is obtained from areas designated by the U. S. Department of Agriculture as modified tuberculosis-free and from cows in herds certified brucellosis-free by the USDA or in the process of being accredited or certified. This determination shall be made by contacting the nearest office of the state officials in charge of animal disease control as listed in the Working Reference of Livestock Regulatory Establishments, Stations and Officials.

4.2.7.1.2 *Establishment sources for military procurement.* Inspection shall be made to determine that all milk is obtained from approved military sources of supply.

4.2.7.2 *Wheat.* Examination and acceptance of this component shall be satisfied by tendering a USDA certificate certifying that the product complies with grade requirements of 3.1.3. Failure to comply with this requirement shall be basis for rejection of the component or rejection of the finished product made therefrom.

4.2.7.3 *Farina, wheat flour, barley malt, malted wheat, sodium chloride, sodium bicarbonate, cocoa and chocolate liquor, sugar vanilla, vanillin, or artificial vanilla flavoring and emulsifiers.* Acceptance of these ingredients for compliance with the requirements of 3.1.4 through 3.1.10, as applicable, shall be based on the examination of the labels, invoices and similar documents. This examination shall be made on each new lot of the component received and not less than once during each contract. A statement of compliance shall be provided by the contractor that the ingredients used conform to the applicable requirements. Noncompliance with one or more requirements shall be cause for rejection of the involved lot and all end items containing a portion of the rejected lot.

4.2.7.4 *Water.* Certified semiannual laboratory test reports from the State or municipality in which the plant is located or the Federal Government, indicating that the water supply is of a safe sanitary quality, shall be available for review by the Government inspector and may be accepted as evidence of compliance with 3.1.11.

4.2.8 *Inprocess examination (contractor inspections).*

4.2.8.1 *Equipment, procedures, and controls.* Examination shall be performed during processing of the milk to determine compliance with 3.2 requirements pertaining to equipment, procedures and controls. Records shall be maintained for specific procedures of time and temperature controls covering product included in the purchase contract. Noncompliance with the requirements of 3.2, as determined by examination of records or actual examination shall be cause for rejection of the involved finished product.

4.2.8.2 *Formulation.* Examination shall be made during the processing of the product to determine conformance with 3.6 limitation of ingredients in the finished product. If product is to be gas packaged, inspection shall be made to assure conformance with the prepackaging requirements of 5.1.1.1. Formulation records shall be maintained. Noncompliance, as determined by examination of the records or actual examination, shall be cause for rejection of the involved finished product.

4.2.9 *Finished product inspection.*

4.2.9.1 *For civil agencies.*

4.2.9.1.1 *Lot size.* An inspection lot shall be one carlot or less, converted to number of primary containers of malted milk to be inspected at one place at one time. More than one production code may be included in the lot if the number of cases from each code represented in the lot is known and the production dates are all within fifteen days of one another.

4.2.9.1.1.1 The sample unit for examinations required in tables II and III is the

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TABLE I. *Single sampling plans*

Number of primary containers in the insp. lot	Minimum sample size	Table II		Minimum sample size	Table III			
					Major		Total	
		AQL 2.5			AQL 1.5		AQL 6.5	
		Ac	Re		Ac	Re	Ac	Re
1—300	9	0	1	6	*		1	2
301—1500	9	0	1	13	0	1	2	3
1501 and over	29	2	3	29	1	2	4	5

* Lot with one or more defect shall be rejected.

contents of one primary container in accordance with table I. The acceptable quality level (AQL) shall be expressed in defects per hundred units.

4.2.9.1.2 Examination of the net weight, product characteristics and interior coating shall be in accordance with the following procedure: The appropriate minimum sample size designated in table I shall be formed by drawing one sample unit per shipping case selected proportionately from the production codes represented in the inspection lot. The applicable acceptance (Ac) and rejection (Re) numbers are given in table I at the intersection of the sample size(s) row and the appropriate AQL column. Classification of defects found during the examination shall be in accordance with tables II and III.

4.2.9.2 *For military agencies.*

4.2.9.2.1 *Examination of the end item.* Classification of defects found during examination shall be in accordance with tables II and III. The examination criteria for the above referenced tables shall be as shown below. Sampling for inspection shall be performed in accordance with MIL-STD-105, except as otherwise stated herein.

Table	Insp. level	Sample unit	Lot size expressed in	AQL'S Defects per hundred units		
				Major	Minor	Total
II	S-4	One primary container	Primary containers	—	2.5	—
III	S-4	One primary container	Primary containers	1.5	—	6.5

4.2.9.2.2 Testing of components described in this specification and changes in the testing of component listed in subsidiary specifications shall be as shown below. Nonconformance to one or more test requirements shall be cause for rejection of the involved finished product made therefrom.

Testing of components

Component	Sample size	Lot size expressed in	Insp. level	Char.	Results reported	Test ref. & reqm. par.
Cans[1]	1 can & lid	cans	S-1	Tinplate	Nearest 0.01 lb. base box	5.1.1 4.3.5

[1] Testing of can coating (when required see 6.3) and labeling shall be accomplished on the same samples used for testing of tinplate. One or more failures for can coating or labeling shall be cause for rejection of the lot.

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TABLE II. *Examination for net weight*[1] *(For civil and military agencies.)*

Category	Defect
Minor	
201	One to ten pounds net weight—2.0 percent or more under specified weight[2]
202	Under one pound net weight—5.0 percent or more under specified weight[3]

[1] Lot shall be rejected if sample data indicates lot average net weight is less than specified net weight.

[2] Report to nearest 1/2 ounce.

[3] Report to nearest 1/8 ounce.

TABLE III. *Examination for product characteristics and interior coating*[1] *(For civil and military agencies.)*

Defect	Product characteristics	Major	Minor
Type or class not as specified		101	
Flavor			
(a) Definite scorched		102	
(b) Slight bitter or slight neutralizer		103	
(c) Slight rancid or slight stale		104	
(d) Other objectionable flavors or odors		105	
Appearance and texture			
(a) Not uniform in color and appearance		106	
(b) Lumps that do not fall apart under slight pressure (classes 1 and 2 only)		107	
Type I product[2]			
(a) Not a pleasing pronounced natural malt flavor		108	
(b) Not a tan to cream color			201
Type II product[2]			
(a) Not a pleasing pronounced chocolate-malt flavor		109	
(b) Not a light to medium chocolate color			202
Type I or II, class 3			
(a) Size not as specified (plus or minus 1/32 inch)			203
Interior coating—cans			
Missing		110	
Blistered or softened areas which can be peeled by fingertip abrasion		111	
Scratches through enamel		112	
Blistered or softened areas which cannot be peeled by fingertip abrasion			204
Denuded areas (other than scratches)			205

[1] Presence of foreign materials shall be the basis for rejection of the entire lot.

[2] Reconstitute products at the rate of approximately 21 g. of product to 250 ml. of cold (45°—55° F.) whole milk.

4.2.9.3 *Condition inspection of containers.* When inspection of product for quality and condition is performed by the U. S. Department of Agriculture, the U. S. Standards for Condition of Food Containers shall apply. (For applicable document governing condition of inspection of containers for military inspection agencies see PPP-C-29.)

4.2.9.3.1 *For civil agencies.* Examination of the exterior condition of the primary containers and labels and markings shall be in accordance with the procedures outlined in U. S. Standards for Condition of Food Containers. (See purchase contract for type and size primary and shipping container and for any special labeling and marking requirements.)

4.2.9.4 *Examination of can labeling.* When can labeling is required to be in accordance with MIL-L-1497, examination of cans for labeling shall be in accordance with that specification.

4.2.9.5 *Examination of shipping container.* When containers are required to be in accordance with PPP-B-636, examination of the filled and closed shipping container shall be in accordance with that specification. In addition, the following defects shall be included in the table of examination: Major: Markings; missing, incorrect, or illegible. Minor: Arrangements not as specified. For level C packing, examination shall be in accordance with the aforementioned specification except that only the defects for marking and arrangement shall apply.

4.2.9.6 *Examination for age of product prior to shipping.* Determination of compliance with requirements of 3.5 shall be ascertained by storage records, or marked date of pack. Noncompliance with the requirements of 3.5 shall be cause for rejection of the lot.

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4.2.10 *For civil agencies—Sampling procedure and acceptance criteria for testing of finished product.* The finished product shall be tested for compliance with analytical requirements of 3.3.2. Procedures for testing shall be in accordance with 4.3.

4.2.10.1 *Contractor testing.* A representative sample of each production code shall be tested. Records of the results shall be kept and be available for review by the Government.

4.2.10.2 *Sampling for verification testing.* The sample size shall be the number of primary containers indicated in table I for examinations under table III, but not more than 13 containers.

4.2.10.3 *Compositing for verification testing.* Sample units may be composited prior to testing for all analytical requirements, except oxygen content. Noncompliance with one or more analytical requirements shall be cause for rejection of the lot.

4.2.11 *For military agencies—Sampling procedure and acceptance criteria for testing of finished product.* The finished product shall be tested for milk, protein, moisture, phosphatase activity, bacterial count, coliform count, oxygen and in addition for class I product only dispersibility performance and density as specified in 3.3.2.1. Procedures for testing shall be in accordance with procedures contained in 4.3 through 4.3.6. Requirements for testing for oxygen shall be applicable to the sample unit. The sample for all other requirements shall be a composite derived by the laboratory from the number of cans tested for oxygen. The sample unit for testing of oxygen shall be one primary container. Lot size shall be expressed in primary containers. Sample size shall be the number of primary containers indicated by inspection level S-2. The AQL shall be 2.5 expressed as defects per hundred units. Results shall be reported to the nearest 0.1 percent. A result not within specified limits shall be a defect. Results for milkfat, protein, and moisture shall be reported to the nearest 0.1 percent. Phosphatase activity shall be reported as pass or fail. Bacterial count shall be reported to the nearest 500 per g. and coliform to the nearest unit. Dispersibility and density shall be reported to the nearest g. Nonconformance to one or more composite requirements shall be cause for rejection of the lot.

4.2.12 *Stand-by samples.* The Government reserves the right to withdraw and hold stand-by test samples of components or finished product or both (quantity of which shall be not more than twice that of the original Government sample) for inspection purposes. These samples, if not used, will be returned to the vendor upon request.

4.3 Tests. Test procedures and controls which differ from those specified herein, may be used by the contractor if they provide a quality assurance equivalent to that specified. If the Government inspection activity determines that such procedures and controls do not provide, as a minimum, such quality assurance, the contractor will use the test procedures set forth herein. In case of dispute as to test results, the test methods specified herein will govern.

4.3.1 *Chemical analyses.* Chemical analyses, unless otherwise specified, if required by purchaser (see 6.3) shall be made in accordance with the following methods from Official Methods of Analysis of the Association of Official Agricultural Chemists:

Test	Source	Method
Milkfat	1	Fat in malted milk.
Protein	1	Total nitrogen (Nx6.38).
Moisture	1	Moisture
Phosphatase activity	2	Residual phosphatase, Method II (reconstitute 13 g. powder per 100 ml. water).

[1] Chapter: Dairy Products, section: Dried Milk and Malted Milk.

[2] Chapter: Dairy Products, section: Milk.

4.3.2 *Sediment and bacteriological analyses.*

4.3.2.1 *Raw milk—(contractor inspection.)* Sediment and bacteriological analyses of raw milk shall be in accordance with 3.1.1.1.

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4.3.2.2 *Finished product.* Bacteriological analyses of the finished product, if required by purchaser (see 6.3) shall be made in accordance with the following methods from Standard Methods for the Examination of Dairy Products, Chapter: Concentrated Milk and Cultured Product, section: Dry Milk:

Test	Method
Bacterial estimate (as standard plate count)	Agar Plate
Coliform count	Coliform Group (Coliform test-solid media using desoxycholate lactose agar)

4.3.3 *Dispersibility performance.* Dispersibility performance, if required by purchaser (see 6.3), shall be determined by the method outlined in Methods of Laboratory Analyses for Dry Whole Milk, Nonfat Dry Milk, Dry Buttermilk and Dry Whey except that the temperature of water shall be 50°F. instead of 72°F.

4.3.4 *Density.* Density, if required by purchaser (see 6.3), shall be determined as follows: Support a powder funnel (8-10 cm. diameter) above a tared 250 ml. graduated cylinder by means of a ring stand in such manner that the end of the funnel is below the top of the cylinder and not touching any part of it. Pour the powder through the funnel into the cylinder until the level of the powder is exactly at the 200 ml. graduation mark, taking care not to jar the cylinder during the filling operation. Weigh the cylinder containing the powder and calculate the weight of the powder. Calculate the density expressed as g. per liter by multiplying the weight of powder by a factory of 5.

4.3.5 *Tin coating weights.* When specified (see 6.3), the tin coating weights shall be determined by any method specified in PPP-C-96.

4.3.6 *Oxygen.* Oxygen content of gas in headspace of one filled can per inspection lot shall be determined in accordance with method 237 of Fed. Std. No. 101. The cans to be tested for oxygen content shall be held at a temperature of not below 60°F. between the time of packing in the final container and the time of testing. The test shall be conducted not less than 7 days after packaging.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The product shall be packaged in accordance with 5.1.1 or 5.1.2, as specified (see 6.3).

5.1.1 *Level A.*

5.1.1.1 *Prepackaging holding requirements (for gas packaged product, see* 6.3). The finished product may be held for not more than 7 days after manufacture before final packaging. If not packaged in the final container within 24 hours after manufacture, the product shall be continually stored either in an atmosphere of inert gas or under a constant vacuum of not less than 28 inches. If the storage container is opened or the vacuum broken more than 24 hours prior to the final packaging, the above procedure shall be repeated. The storage containers shall not have surfaces of iron or copper exposed to the product.

5.1.1.2 Eight ounces of the class 1 product (based on minimum density requirement of 3.3.2.1.2) or 10 ounces of the class 2 product shall be filled into a size 303 by 406 open-top style, round metal can with soldered side seam and compound-lined, double-seamed ends. The cans shall be made throughout from not less than commercial 0.25-pound per base box electrolytic tinplate (see 4.3.5) and shall be coated outside with a coating conforming to type I or II or when specified (see 6.3) type III, of TT-C-495. The cans shall be coated overall inside with an enamel furnished by the can manufacturer in accordance with the best commercial practice for the product. The cans shall be hermetically sealed. The product shall be gas-packed by first removing the air from the filled can and replacing it with nitrogen. If tip soldering is used to close the

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exhaust vents for nitrogen filling, a suitable noncorrosive flux shall be used. When the ends of the can are clinched prior to vacuumizing and gassing, the flange ends should be indented or dimpled to prevent seal off trouble during the nitrogen flush. The oxygen content of the gases in the sealed container, not less than 7 days after filling, shall not exceed 2.0 percent when tested as specified in 4.3.6.

5.1.2 *Level B (civil agencies).* Five pounds of the product shall be packaged in a square, rectangular, or round metal can with soldered side seam and compound-lined, double-seamed ends. The top shall be fitted with a screw-neck opening. The closure shall consist of a pry-out inner seal and curled-edge knurled screw cap. The cans shall be made throughout from commercial 0.25 pound electrolytic tinplate. The filling processes shall conform to established commercial standards for sanitation.

5.1.3 *Level C.*

5.1.3.1 *Cans, military agencies.* The product shall be packaged as specified in 5.1.1.2 except that cans with or without a commercial exterior coating will be acceptable.

5.1.3.2 *Cans or glass, civil agencies.* Unless otherwise specified (see 6.3), commercial packaging is acceptable. No material shall be used in packaging and packing of the product which will impart an odor or flavor to the product. All containers shall be sound and clean and shall adequately protect the contents from contamination.

5.2 Packing. The product shall be packed in accordance with 5.2.1, 5.2.2, or 5.2.3 as specified (see 6.3).

5.2.1 *Level A.* Twenty-four cans of the product, arranged 4 in length, 3 in width, and 2 in depth, shall be packed on end in a snug-fitting fiberboard box, with sleeve, constructed, closed, and metal strapped in accordance with style RSC-S1, V2s of PPP-B-636.

5.2.2 *Level B.* Twenty-four cans of the product, arranged as specified in 5.2.1, shall be packed in a snug-fitting fiberboard box constructed and closed in accordance with style RSC, type CF or SF, class domestic of PPP-B-636.

5.2.2.1 When specified (see 6.3 and 6.4), the container shall be a fiberboard box constructed, closed, and reinforced in accordance with style RSC, V3c or V3s of PPP-B-636.

5.2.3 *Level C.*

5.2.3.1 *Military agencies.* Twenty-four cans of the product, arranged as specified in 5.2.1, shall be packed in containers which are acceptable by common carriers for safe transportation to the point of destination specified in shipping instructions at the lowest transportation rate for such supplies.

5.2.3.2 *Civil agencies.* The product, packaged as specified in 5.1.3.2, shall be packed in commercial containers as specified in 5.2.3.1.

5.3 Labeling and marking.

5.3.1 *Civil agencies.*

5.3.1.1 *Unit container.* Any commercial labeling or additional labeling as specified, that complies with Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder, will be acceptable. Labeling shall include the name of the product; name and address of manufacturer, packer or distributor; a statement of net content and a list of ingredients where necessary.

5.3.1.2 *Shipping containers.* Unless otherwise specified, shipping containers shall be marked in accordance with Fed. Std. No. 123 and such other pertinent information as specified by individual agencies.

5.3.2 *Military agencies.*

5.3.2.1 *Cans.* Cans shall be labeled in accordance with MIL-L-1497 and as follows:

DIRECTIONS FOR USE

Add one ounce of dry product to 8 fluid

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ounces of either hot or cold milk. Stir and serve.

5.3.2.2 *Shipping containers.* Shipping container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use of product, and technical information. The dry malted milk covered by this specification is intended as an ingredient in other food products, or reconstituted as a beverage. Class 1 product is more bulky than the class 2 but is more easily reconstituted in cold water or milk. Malted milk has a high nutritive value, easily digestible and requires a small amount of shipping space in comparison to fluid products. It is highly hygroscopies; therefore, it must not be stored in areas or in open or damaged containers which will allow moisture pick-up in the product. Exposure to ordinary atmosphere (especially high humidity) for as short a period as overnight will inevitably cause the product to become moist, soft, and sticky. When kept in the sealed container, the product is classified as nonperishable and has long usable shelf-life even under unfavorable storage conditions. Storage in a cool location will a d in preserving a highly acceptable flavor. Packaging in an atmosphere of air (normal commercial packaging) is normally satisfactory; however, packaging in an atmosphere of inert gas (reduced oxygen) will greatly extend shelf-life if the product is held at high temperatures (see 6.2).

6.2 Information for gas packaging requirements. Necessity for gas packaging (reduced oxygen content in container) is dependent upon storage and condition of product. When packaged under an atmosphere of air, the product should not be stored for a period more than 3 months if the storage temperature is at a constant temperature of 100°F.; more than one year at normal United States ambient temperatures, nor for a period which will include two summers. If it is anticipated that these conditions will not be violated, packaging in hermetically sealed containers (atmosphere of air) should be specified, otherwise packaging under an atmosphere of inert gas in hermetically sealed containers should be required.

6.3 Ordering data. Purchasers should select the preferred options permitted herein and incude the following information in procurement documents:

(*a*) Title, number, and date of this specification.
(*b*) Type and class of product required (see 1.2).
(*c*) When product shall be shipped (see 3.4).
(*d*) When the supplier is not required to perform the examination and tests (see 4.1).
(*e*) Where final inspection shall take place if not at the point of inspection (see 4.2.4).
(*f*) When USDA certificates of quality and condition are required (see 4.2.5).
(*g*) When verification testing for analytical, in process, and component examinations may be waived on small lot orders (see 4.2.5).
(*h*) When chemical analyses are required (see 4.3.1).
(*i*) When bacterial analyses are required (see 4.3.2.2).
(*j*) When dispersibility performance tests are required (see 4.3.3).
(*k*) When density tests are required (see 4.3.4).
(*l*) When tin coating weight tests are required (see 4.2.9.2.2 and 4.3.5).
(*m*) Levels of packaging and packing required (see 5.1 and 5.2).
(*n*) When gas packaging is required (see 5.1.1.2 and 6.2).
(*o*) When other than commercial packaging is required for civil agencies (see 5.1.3.2).
(*p*) When V3c or V3s, class domestic, style RSC fiberboard box is re-

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quired for level B packing (see 5.2.2.1).

(q) When shipment prior to receipt of analyses by government laboratory is permitted.

6.4 Packing specified in 5.2.2.1 is intended to provide the limited military protection required for shipment of supplies to certain specified overseas destinations; transfer at sea; and for shipment and storage of Special Program Reserves.

6.5 **Contractor's inspection.** The contractor should be required to assure that the product conforms with the specification prior to submission to the Government for final inspection.

6.6 **Destination inspection.** When the finished product has been inspected and passed at point other than destination, the contract should require that the product be inspected at destination for cond.tion and quantity only. Unless otherwise specified, such inspection should be made by the receiving facility. However, inspection for quality may be made by the procuring agency or a duly authorized representative, when deemed necessary to verify contract compliance.

6.7 **Transportation descriptions.** Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Milk, malted, in barrels, boxes, paper lined bags, or in cans.
Carload minimum weight 36,000 pounds.

Motor:

Milk, malted, in barrels, boxes, or cans.
Truckload minimum weight 30,000 pounds, subject to Rule 115, National Motor Freight Classification.

MILITARY INTEREST:

Custodians:

Army—GL
Navy—SA
Air Force—67

Review Activities:

Army—MD
Navy—MC, SA
Air Force—67

Review activities information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification List for DOD Standardization Documents.

CIVIL AGENCY INTEREST:

Preparing activity:

AGR—C & MS

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POSTAGE AND FEES PAID
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HEAD, PROCUREMENT MANAGEMENT DIVISION
OFFICE OF PLANT & OPERATIONS
DEPARTMENT OF AGRICULTURE
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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